To: Board of Education
From: Dr. Jon Franco, Senior Chief of Operations
Cc: Dr. Kimberlee Armstrong, Superintendent
Deborah Kafoury, Chief of Staff
RE: Seismic Memo
Date: March 10, 2026

Resolution 7109 establishes the criteria and intent for prioritizing seismic retrofits. It contemplates that all high-risk schools will not be addressed within a single bond cycle. Specifically, it directs Portland Public Schools (PPS) to use current seismic assessment data to identify and prioritize schools that pose the greatest risk of injury or death in a significant earthquake, and to clearly communicate the criteria used to determine the sequence of retrofits.

We acknowledge that discussion of a significant earthquake can cause concern and anxiety. We affirm at all times that we are focused on ensuring the safety of all of our children and staff in our buildings and are working to devise a plan to prioritize the work across our many buildings to do that.

Holmes and PPS Partnership

Holmes was engaged by PPS to:

- Complete a district-wide seismic risk assessment of PPS school facilities, and
- Use those results to help PPS identify a first tranche of seismic retrofit projects that could be delivered with up to approximately 100 million dollars of 2025 bond funding for 8-10 schools dedicated to addressing high seismic risks with the District[1].

This memo summarizes the technical approach Holmes used, in partnership with PPS, to recommend schools and project types for this first phase of work, and how that approach aligns with Resolution 7109 including:

1. Metrics and Formula Used to Identify Retrofit Candidates
2. Seismic and facility metrics

Identification of Schools

Holmes and PPS explored several options combining available data into ranking formulas resulting in the identification and prioritization of schools for seismic retrofits. The metrics utilized in these formulas include the following:

[1] *The budget for seismic retrofits from the 2025 bond is principally created by savings on other 2025 bond projects. PPS continues to track progress towards those savings goals and accelerate the seismic retrofit work whenever possible.*



Seismic Risk Score (Holmes)
Each distinct "building part" on a campus was assigned a Seismic Risk Score from 1 to 10, where higher scores indicate poorer expected performance and greater life-safety risk in an earthquake. Scores were based on building type, construction era, and observed or documented seismic vulnerabilities.

Deferred Maintenance Score (PPS)
A 0–5 score (higher means more need) representing overall facility condition and backlog of deferred maintenance work. This score is relevant to seismic retrofit prioritization because it reflects the extent of prior structural, envelope and roof investments made at each campus. ***The Deferred Maintenance score also incorporates an equity adjustment recognizing CSI/TSI status.***

Enrollment / "Right-sizing" Indicator (PPS)
A simple Yes/No indicator of whether a school's enrollment reflects the number of students who would benefit from improved seismic performance using a threshold of 300 students. Given the rightsizing process that we will be engaging in, we assessed the projected enrollment of our school sites over the next 3 years. Schools not meeting the enrollment threshold of 300 were not prioritized for this first round of recommendations.

Because these measures use different scoring scales, we first convert the Seismic Risk and Deferred Maintenance scores to a common 0-to-1 scale, and treat Yes/No answers (like Enrollment) as 1 or 0. This lets us combine them fairly in one formula so that the stated weights (90 percent seismic, 5 percent maintenance, 5 percent enrollment) accurately reflect their influence.

Final recommended formula

Holmes developed and tested several weighting formulas to see how campus rankings changed when different metrics were emphasised. This sensitivity testing allowed PPS and Holmes to confirm which schools consistently appeared as high priorities when seismic risk was emphasized, and to understand how rankings would shift if non-seismic factors were given more weight.

After review with PPS, the following formula was recommended as the primary technical tool for identifying high-priority full campus retrofits:

$$Score = 0.90\times(Highest\ Campus\ Seismic\ Risk\ Score) + 0.05\times(Deferred\ Maintenance) + 0.05\times(Enrollment)$$

In practical terms, this formula prioritizes campuses with the greatest potential for structural failure or collapse in an earthquake, while modestly accounting for facility condition and the number of students served.

This structure reflects Resolution 7109's direction to prioritize schools that "pose the greatest risk of injury or death in a significant seismic event", while recognizing that modestly combining seismic work with existing facility needs and serving more students can provide additional benefit.



Interpreting the formula

Across the different scenarios, a consistent group of high-risk campuses repeatedly appeared near the top of the rankings. In particular (in alphabetical order), Fernwood (Beverly Cleary), Glencoe, Grout, Jackson, Kelly, Rose City Park, and Roseway Heights appeared in the top ten for more than half of the formulas that were tested. These patterns gave PPS and Holmes confidence that the formulas were identifying a common core of high-risk schools, even when non-seismic factors were varied.

The final recommended formula was then applied to all campuses to generate a primary ranked list. This ranked list was not treated as a strict “top N” selection. Instead, it was used to:

- Highlight a manageable group of highest-scoring campuses for further review, and
- Provide a consistent, data-driven starting point for conversations about scope, cost, and phasing.

Joint discussions between Holmes and PPS

Holmes worked with PPS facilities and operations staff to refine the shortlist to a program that:

- Would have the **greatest impact on reducing life-safety risk** across the overall building stock, in keeping with Resolution 7109, and
- Could realistically be delivered within approximately 100 million dollars and the Board’s direction to address **8–10 schools** in this first phase.

Key considerations in those discussions included:

- The type and extent of high-risk structural deficiencies at each campus,
- The likely rough order of magnitude (ROM) cost to achieve a meaningful seismic improvement,
- The practicality of combining or staging work, and
- The need to balance a small number of full campus projects with a larger number of targeted interventions that collectively protect more students.

Some high-risk campuses would require very large investments to achieve a full seismic retrofit. In these cases, PPS concluded, in consultation with Holmes, that more detailed investigation and scope refinement, along with long-range facilities planning, should occur before committing a substantial portion of this first bond allocation. These campuses remain high priorities for future funding. Sequencing reflects affordability and deliverability constraints within this bond cycle, not a diminished commitment to the safety of the students served in those buildings.

For other campuses, the data showed that targeted Unreinforced Masonry (URM) or building-part retrofits could produce a large reduction in life safety risk at significantly lower cost than a full campus retrofit. These opportunities became key components of the recommended program.

In interpreting these results, it was also important to note how “building parts” were defined. The underlying seismic assessments (Phase 1) were scoped and completed before PPS decided to use them



for district-wide risk scoring (Phase 2). Holmes engineers grouped spaces into building parts based on available drawings and site observations. In some cases, a single building part encompasses a large floor area, and a more granular subdivision would more clearly distinguish the different risk characteristics of the spaces. Rather than re-segment every campus, Holmes and PPS relied on our detailed familiarity with each school to recognize where this coarser grouping might either mask a high-risk area or make an entire campus appear higher-risk when only part of a building is driving the score. For example, a gymnasium, classroom wing, and administration area may be grouped together for scoring purposes even though their seismic vulnerabilities differ, requiring engineering judgment when interpreting campus-level scores.

With that shared understanding of the data and its limitations, Holmes and PPS shaped a package of projects that fits within the available anticipated budget to reduce seismic risk across the district.

Types of Projects Recommended

The resulting program under the 2025 bond includes three main project types, each linked to the risk metrics:

1. **Full campus retrofits**
    - Applied at the **highest-priority campuses per the seismic formula** and engineering review, where multiple building parts exhibit very high Seismic Risk Scores.
    - Full campus projects provide comprehensive risk reduction for those schools, addressing multiple interacting deficiencies.
2. **Targeted retrofits – high-risk building parts**
    - Focused on **specific buildings or building parts** that contribute heavily to a campus's Seismic Risk Score.
    - These projects were chosen where strengthening those elements will **significantly reduce both part-level and campus-level seismic risk**, for example, URM gymnasiums or classroom wings.
3. **Targeted retrofits – localized URM at egress or high-occupancy areas**
    - Focused on **local URM walls or components** near main egress routes, gyms, auditoriums, and other high-occupancy spaces.
    - Even when these localized elements do not substantially change the overall Seismic Risk Score for the campus, strengthening them provides strong "value for money" from a life safety perspective, because they directly reduce the risk in spaces where many students and staff may be present during an event. These items have been previously identified in the Holmes report table in the "Small Project Opportunities, Relative Low Cost to High Safety Return".

Across all project types, the primary objective is to reduce the likelihood of serious injury or loss of life during a seismic event, particularly in spaces where students and staff are most likely to be present.



First Phase within a Longer-Term Program

Resolution 7109 recognizes that the seismic risk within PPS cannot be fully addressed in a single bond cycle. The 2025 bond allocation for seismic work represents a **first phase**, focusing on the highest-risk buildings and the most cost-effective risk-reduction opportunities identified by the current assessment. Exclusion from this initial phase does not indicate lower priority; rather, it reflects sequencing, affordability, and deliverability constraints within this bond cycle.

Holmes recommends that:

- The Seismic Risk Scores and URM information developed in this study continue to be used as a **primary technical tool** for future bond planning, grant applications (such as the State's Seismic Rehabilitation Grant Program), and capital programming.

- Schools that were identified as high-risk but not advanced in this bond cycle, particularly those with complex or high-cost retrofit needs, be **prioritized for further investigation and future planning and funding**.

In this way, PPS can continue to act on the commitments in Resolution 7109: using transparent, data-driven criteria, focusing first on the highest-risk buildings, and advancing a multi-bond, multi-year strategy to bring all high-risk schools to an acceptable standard of seismic safety.